

Lab 7

Part A

```
import nltk

nltk.download('wordnet')

[nltk_data] Downloading package wordnet to /root/nltk_data...
True

from nltk.corpus import wordnet as wn

wn.synsets('sink')

[Synset('sink.n.01'),
 Synset('sink.n.02'),
 Synset('sinkhole.n.01'),
 Synset('cesspool.n.01'),
 Synset('sink.v.01'),
 Synset('sink.v.02'),
 Synset('sink.v.03'),
 Synset('sink.v.04'),
 Synset('sink.v.05'),
 Synset('dip.v.08'),
 Synset('slump.v.03'),
 Synset('slump.v.02'),
 Synset('bury.v.05')]

print(len(wn.synsets('sink')))

13

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```

The number of synsets for the english word 'sink' is 13. From listing out the synsets of 'sink', we see that it has two different parts of speech 'n' for noun and 'v' for verb.

```
sink = wn.synset('sink.n.01')

sink

Synset('sink.n.01')

sink.hypernyms()

[Synset('plumbing_fixture.n.01')]
```

plumbing fixture is the direct hypernym of the most common noun sense of sink

```
drink = wn.synset('drink.v.01')

drink.hyponyms()

[Synset('gulp.v.01'),
 Synset('guzzle.v.01'),
 Synset('toss_off.v.02'),
 Synset('drain_the_cup.v.01'),
 Synset('guggle.v.03'),
 Synset('suck.v.01'),
 Synset('lap.v.04'),
 Synset('sip.v.01'),
 Synset('swill.v.02')]
```

The direct troponyms of drink are shown above

```
dog = wn.synset('dog.n.01')

insect = wn.synset('insect.n.01')

dog.lowest_common_hypernyms(insect)

[Synset('animal.n.01')]
```

The closest common ancestor in the IS-A hierarchy for dog.n.01 and insect.n.01 is animal.n.01

```
astronaut = wn.synset('astronaut.n.01')

astronaut.instance_hyponyms()

[Synset('gagarin.n.01'),
 Synset('armstrong.n.01'),
 Synset('glenn.n.01'),
 Synset('shepard.n.02'),
 Synset('tereshkova.n.01')]

len(astronaut.instance_hyponyms())

5
```

There are 5 named astronauts which are instances of astronaut.n.01

## Part B

```
import re
```

```
from google.colab import drive
gdrive_mount = '/content/drive'
drive.mount(gdrive_mount, force_remount=True)
```

```
Mounted at /content/drive
```

```
labdir = '/content/drive/My Drive/Teaching/605.646/Fall 2025/lab7/'
%cd "$labdir"
```

```
/content/drive/My Drive/Teaching/605.646/Fall 2025/lab7
```

```
with open('wiki.train.txt','r',encoding='utf-8') as f:
    trainLines = f.readlines()
```

```
trainLines[0:10]
```

```
['sandpaper\tabrasive\tIt is commonly used as an abrasive, on everything from sandpaper to large machines used in machining metals, plastics, and wood.\n',
 'bag\taccessory\t- Accessory bag with the following items.\n',
 'bag\taccessory\tWhen Julie feels unable to bide her time for a particular fashion accessory, such as a Hermès Birkin bag, for which there is a three-year wait, she shoplifts from the family store - "kind of stealing from yourself".\n',
 'coincidence\taccident\tIn a coincidence, Ambujam's bullock-cart and another car strike each other in a road accident, right in front of the very same police station.\n',
 'coincidence\taccident\tThe role and legitimacy of coincidence has frequently been the topic of heated arguments ever since Ronald A. Knox categorically stated that "no accident must ever help the detective" (Commandment No. 6 in his "Decalogue").\n',
 'concussion\taccident\tDuring an examination at the local hospital the doctor comes to the conclusion that Cassie only has some scratches and not even a concussion, but Cassie has lost her memory due to the accident.\n',
 'concussion\taccident\tHe suffered his fourth severe concussion in six months and had a compound fracture of his right humerus, the same arm that was nearly amputated when he was 14 due to a hay equipment accident.\n',
 'concussion\taccident\tMalika was rendered unconscious by the collision, but survived the accident with a mild concussion.\n',
 'concussion\taccident\tOn the day of the crash, her teeth were removed from the back of the roof of her mouth and moved forward without anesthesia due to the concussion she suffered in the accident.\n',
 'concussion\taccident\tRita begins to notice a pattern of lies from Dexter, including deceiving her about the concussion he suffered in the car accident and the ownership of his old apartment.\n']
```

```
trainData = []
for line in trainLines:
    line = line.strip()
    if not line:
        continue
    parts = line.split('\t')
    if len(parts) >= 3:
        hyponym, hypernym, sentence = parts[0], parts[1], parts[2]
        trainData.append((hyponym, hypernym, sentence))
```

```
trainData[0:10]
```

```
[('sandpaper',
 'abrasive',
 'It is commonly used as an abrasive, on everything from sandpaper to large machines used in machining metals, plastics, and wood.'),
 ('bag', 'accessory', '- Accessory bag with the following items.'),
 ('bag',
 'accessory',
 'When Julie feels unable to bide her time for a particular fashion accessory, such as a Hermès Birkin bag, for which there is a three-year wait, she shoplifts from the family store - "kind of stealing from yourself".'),
 ('coincidence',
 'accident',
 'In a coincidence, Ambujam's bullock-cart and another car strike each other in a road accident, right in front of the very same police station.'),
 ('coincidence',
 'accident',
 'The role and legitimacy of coincidence has frequently been the topic of heated arguments ever since Ronald A. Knox categorically stated that "no accident must ever help the detective" (Commandment No. 6 in his "Decalogue").'),
 ('concussion',
 'accident',
 'During an examination at the local hospital the doctor comes to the conclusion that Cassie only has some scratches and not even a concussion, but Cassie has lost her memory due to the accident.'),
 ('concussion',
 'accident',
 'He suffered his fourth severe concussion in six months and had a compound fracture of his right humerus, the same arm that was nearly amputated when he was 14 due to a hay equipment accident.'),
 ('concussion',
 'accident',
 'Malika was rendered unconscious by the collision, but survived the accident with a mild concussion.'),
 ('concussion',
 'accident',
 'On the day of the crash, her teeth were removed from the back of the roof of her mouth and moved forward without anesthesia due to the concussion she suffered in the accident.'),
 ('concussion',
 'accident',
 'Rita begins to notice a pattern of lies from Dexter, including deceiving her about the concussion he suffered in the car accident and the ownership of his old apartment.')] 
```

```
pattern1 = re.compile(r'(?P<hypernym>[\w\s]+?)\s+such\s+as\s+(?P<hyponyms>[\w\s,]+(?:\s+(?:and|or)\s+[\w\s]+)?)', re.IGNORECASE)
```

```
pattern2 = re.compile(r'(?P<hypernym>[\w\s]+?)\s*,\s*\s+including\s+(?P<hyponyms>[\w\s,]+(?:\s+(?:and|or)\s+[\w\s]+)?)', re.IGNORECASE)
```

```
Start coding or generate with AI.
```

```
trainSentences = [sent for i, j, sent in trainData]
```

```
def parse_hyponyms(hyponymString):
    hyponymString = re.sub(r'\s+and\s+', ',', hyponymString, flags=re.IGNORECASE)
    hyponymString = re.sub(r'\s+or\s+', ',', hyponymString, flags=re.IGNORECASE)
    hyponyms = [h.strip() for h in hyponymString.split(',') if h.strip()]
    hyponyms = [h for h in hyponyms if len(h)<50]
    return hyponyms
```

```
patterns = [{"such as": pattern1}, {"including": pattern2}]
```

```
relations = {"such as": [], "including": []}
```

```
for sentence in trainSentences:
    sentence = sentence.strip()
    if not sentence:
```

continue

```
for pattern_name, pattern in patterns:
    matches = pattern.finditer(sentence)

    for match in matches:
        hypernym= match.group('hypernym').strip()
        hyponyms_str=match.group('hyponyms').strip()
        hypernym= re.sub(r'[;,!?]+\$', '', hypernym).strip()
        hyponyms=parse_hyponyms(hyponyms_str)

        if hypernym and len(hypernym)<30:
            for hyponym in hyponyms:
                if hyponym and len(hyponym)<30:
                    relations[pattern_name].append((hyponym,hypernym,sentence))
```

len(relations)

2

relations["such as"][0:20]

```
[('the Airbus A340',
'Aircraft',
'Aircraft such as the Airbus A340 and the Boeing 747-400 use winglets.'),
('the Boeing 747',
'Aircraft',
'Aircraft such as the Airbus A340 and the Boeing 747-400 use winglets.'),
('the locomotive',
'based vehicles',
'Consequently, ground-based vehicles such as the locomotive and automobile were replaced by aircraft such as the airplane and the zeppelin as the leading mode of transportation in North America.'),
('James Stewart',
'including Hollywood notables',
'After the war Connelly and Hayward raised $2,000,000 (in 1946 dollars) from investors, including Hollywood notables such as James Stewart and Darryl Zanuck, to expand Southwest into the airline business, pending government approval.'),
('Darryl Zanuck',
'including Hollywood notables',
'After the war Connelly and Hayward raised $2,000,000 (in 1946 dollars) from investors, including Hollywood notables such as James Stewart and Darryl Zanuck, to expand Southwest into the airline business, pending government approval.'),
('pending government approval',
'including Hollywood notables',
'After the war Connelly and Hayward raised $2,000,000 (in 1946 dollars) from investors, including Hollywood notables such as James Stewart and Darryl Zanuck, to expand Southwest into the airline business, pending government approval.'),
('povidone',
'a virucidal antiseptic',
'If available, a virucidal antiseptic such as povidone-iodine, iodine tincture, aqueous iodine solution, or alcohol (ethanol) should be applied after washing.'),
('brass',
'or a suitable alloy',
'Gas checks are most commonly found in the form of a thin cup or disc made of copper, zinc, aluminum, or a suitable alloy such as brass.'),
('moose',
'supporting animal species',
'Cold coniferous forests predominate in the north of the country, supporting animal species such as moose and the Asian black bear, along with over 120 bird species.'),
('the Asian black bear',
'supporting animal species',
'Cold coniferous forests predominate in the north of the country, supporting animal species such as moose and the Asian black bear, along with over 120 bird species.'),
('camels',
'birds species',
'Animal & Birds: Deyr county has variant animals & birds species such as camels, deer, foxes, rabbits, owls, hawks & Flamingo, while caracals, striped hyenas, wolves & Jackels rarely can be sighted.'),
('deer',
'birds species',
'Animal & Birds: Deyr county has variant animals & birds species such as camels, deer, foxes, rabbits, owls, hawks & Flamingo, while caracals, striped hyenas, wolves & Jackels rarely can be sighted.'),
('foxes',
'birds species',
'Animal & Birds: Deyr county has variant animals & birds species such as camels, deer, foxes, rabbits, owls, hawks & Flamingo, while caracals, striped hyenas, wolves & Jackels rarely can be sighted.'),
('rabbits',
'birds species',
'Animal & Birds: Deyr county has variant animals & birds species such as camels, deer, foxes, rabbits, owls, hawks & Flamingo, while caracals, striped hyenas, wolves & Jackels rarely can be sighted.'),
('owls',
'birds species',
'Animal & Birds: Deyr county has variant animals & birds species such as camels, deer, foxes, rabbits, owls, hawks & Flamingo, while caracals, striped hyenas, wolves & Jackels rarely can be sighted.'),
('hawks',
'birds species',
'Animal & Birds: Deyr county has variant animals & birds species such as camels, deer, foxes, rabbits, owls, hawks & Flamingo, while caracals, striped hyenas, wolves & Jackels rarely can be sighted.'),
```

From the 'such as' pattern we can see from the above it gets 'the Boeing 747' as a correct hyponym of 'Aircraft'. An incorrect one is 'foxes' and 'bird species'

relations["including"][0:20]

```
[('associate Dean of Arts',
'serving in various positions',
'He distinguished himself as a teacher and administrator at the University of Waterloo, serving in various positions including associate Dean of Arts.'),
('the Senior Tutor',
'eleven members of the SCR',
'It comprises the Principal; eleven members of the SCR including the Senior Tutor, the Dean, the College Administrator and the College Residence Officer; and eleven members of the JCR including the JCR executive.'),
('the Dean',
'eleven members of the SCR',
'It comprises the Principal; eleven members of the SCR including the Senior Tutor, the Dean, the College Administrator and the College Residence Officer; and eleven members of the JCR including the JCR executive.'),
('the College Administrator',
'eleven members of the SCR',
'It comprises the Principal; eleven members of the SCR including the Senior Tutor, the Dean, the College Administrator and the College Residence Officer; and eleven members of the JCR including the JCR executive.'),
('the JCR executive',
'and eleven members of the JCR',
'It comprises the Principal; eleven members of the SCR including the Senior Tutor, the Dean, the College Administrator and the College Residence Officer; and eleven members of the JCR including the JCR executive.'),
('agriculture',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management, animal management and sciences, sports fisheries and countryside management.'),
('horticulture',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
```

```
animal management and sciences, sports fisheries and countryside management.'),
('medicinal horticulture',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
animal management and sciences, sports fisheries and countryside management.'),
('landscape management',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
animal management and sciences, sports fisheries and countryside management.'),
('garden design',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
animal management and sciences, sports fisheries and countryside management.'),
('equine management',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
animal management and sciences, sports fisheries and countryside management.'),
('animal management',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
animal management and sciences, sports fisheries and countryside management.'),
('sciences',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
animal management and sciences, sports fisheries and countryside management.'),
('sports fisheries',
'based training',
'It is concerned with a wide range of land-based training including agriculture, horticulture, medicinal horticulture, landscape management, garden design, equine management,
```

From the 'including' pattern, a correct example is 'the Senior Tutor' and 'eleven members of the SCR'. An incorrect example is the 'agriculture' and 'based training'. It should be 'land-based training'.

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```
with open('wiki.test.txt','r',encoding='utf-8') as f:
    testLines=f.readlines()
```

```
testData= []
```

```
for line in testLines:
    line = line.strip()
    if not line:
        continue
    parts = line.split('\t')
    if len(parts) >= 3:
        testData.append((parts[0],parts[1],parts[2]))
```

```
testData[0:10]
```

```
[('stillbirth',
'abortion',
'Doctors had expected a stillbirth and recommended an abortion, even though illegal in the Phillipines, to protect her life, but she decided not to have one.'),
('stillbirth',
'abortion',
'Statues of the Boddhisattva Jizo, erected in memory of an abortion, miscarriage, stillbirth, or young childhood death, began appearing at least as early as 1710 at a temple in
Yokohama (see religion and abortion).'),
('richness',
'abundance',
'"Richness' refers to an abundance of such resources."),
('richness',
'abundance',
'First, the relative abundance of ethynyl is an indication of the carbon-richness of its environment (as opposed to oxygen, which provides an important destruction
mechanism).'),
('richness',
'abundance',
'It considers how such interactions, along with interactions between species and the abiotic environment, affect community structure and species richness, diversity and
patterns of abundance.'),
('richness',
'abundance',
'Richness can also refer to at least basic needs being met with abundance widely shared.'),
('richness',
'abundance',
'The abundance of limestone and richness in rainfall in the area has led to a very rich karstic relief, with many caves with prehistoric remains of human occupation.'),
('richness',
'abundance',
'The richness and abundance of these combined 'base economies' grew to such extent that Castro Verde became a regional centre of commerce and thus a crossroad of cultures
within the Mediterranean area."),
('richness',
'abundance',
'They include Species richness, abundance, and Shannon index.'),
('betatron',
'accelerator',
'Another type of circular accelerator, invented in 1940 for accelerating electrons, is the Betatron, a concept which originates ultimately from Norwegian-German scientist Rolf
Widerøe.')]
```

```
testSentences = [sent for i, j, sent in testData]
```

```
testRelations = {"such as": [], "including":[]}
```

```
for sentence in testSentences:
    sentence = sentence.strip()
    if not sentence:
        continue
    for pattern_name, pattern in patterns:
        matches = pattern.finditer(sentence)

        for match in matches:
            hypernym =match.group('hypernym').strip()
            hyponyms_str= match.group('hyponyms').strip()
            hypernym=re.sub(r'[.,:!?]+$', '', hypernym).strip()
            hyponyms=parse_hyponyms(hyponyms_str)

            if hypernym and len(hypernym)<30:
                for hyponym in hyponyms:
                    if hyponym and len(hyponym)<30:
                        testRelations[pattern_name].append((hyponym,hypernym,sentence))
```

```
testRelations["such as"][0:10]
```

```
[[('an Epinephrine pen',
  'Frequently medications',
  'Frequently medications such as an Epinephrine pen or an Antihistamine such as Diphenhydramine (Benadryl) are prescribed by an allergist in case of accidental ingestion.'),
 ('Benadryl',
  'and an antihistamine',
  'In cases of anaphylaxis, emergency medical personnel typically administer epinephrine (available as an autoinjector, such as EpiPen) and an antihistamine such as Benadryl (diphenhydramine).'),
 ('Cézanne',
  'based modernist painters',
  'Under the influence of Chinese artist and art teacher Liu Haisu (1896–1994), Liu admired, and often appropriated the styles of French-based modernist painters such as Cézanne, van Gogh and Matisse.'),
 ('van Gogh',
  'based modernist painters',
  'Under the influence of Chinese artist and art teacher Liu Haisu (1896–1994), Liu admired, and often appropriated the styles of French-based modernist painters such as Cézanne, van Gogh and Matisse.'),
 ('Matisse',
  'based modernist painters',
  'Under the influence of Chinese artist and art teacher Liu Haisu (1896–1994), Liu admired, and often appropriated the styles of French-based modernist painters such as Cézanne, van Gogh and Matisse.'),
 ('Rough Trade Records',
  'as association of companies',
  'The situation improved further with the establishment of 'The Cartel', as association of companies such as Rough Trade Records, Backs Records, and Red Rhino, who helped to take releases from small labels and get them into the shops nationwide.'),
 ('Backs Records',
  'as association of companies',
  'The situation improved further with the establishment of 'The Cartel', as association of companies such as Rough Trade Records, Backs Records, and Red Rhino, who helped to take releases from small labels and get them into the shops nationwide.'),
 ('Red Rhino',
  'as association of companies',
  'The situation improved further with the establishment of 'The Cartel', as association of companies such as Rough Trade Records, Backs Records, and Red Rhino, who helped to take releases from small labels and get them into the shops nationwide.'),
 ('Dore Gold',
  'Critics',
  'Critics such as Dore Gold, Alan Dershowitz, Mark Dreyfus, Robert S. Wistrich, Alan Keyes, and the Anti-Defamation League consider UN attention on Israel's treatment of Palestinians to be excessive.'),
 ('Alan Dershowitz',
  'Critics',
  'Critics such as Dore Gold, Alan Dershowitz, Mark Dreyfus, Robert S. Wistrich, Alan Keyes, and the Anti-Defamation League consider UN attention on Israel's treatment of Palestinians to be excessive.')]]
```

Double-click (or enter) to edit

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